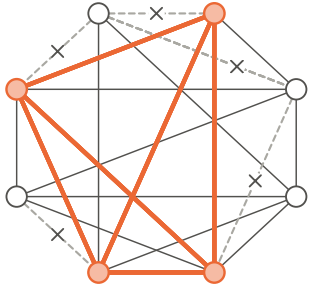
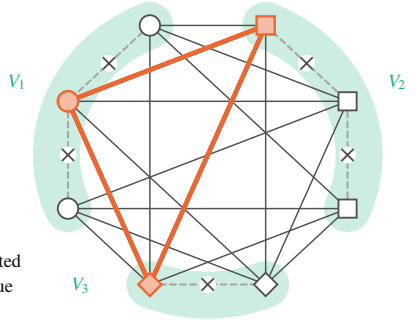
(a) 5 edges deleted; largest clique left: 4



deleted

clique

(b) why the optimum is 3



Leader: delete the 5 edges inside the 3 groups. What is left is 3-partite, so no 4-clique. Upper bound: $\omega \leq 3$.

Follower: any 5 deletions leave $17 > \lfloor 8^2/4 \rfloor = 16$ edges, so a triangle survives (Mantel). Lower bound: $\omega \geq 3$.